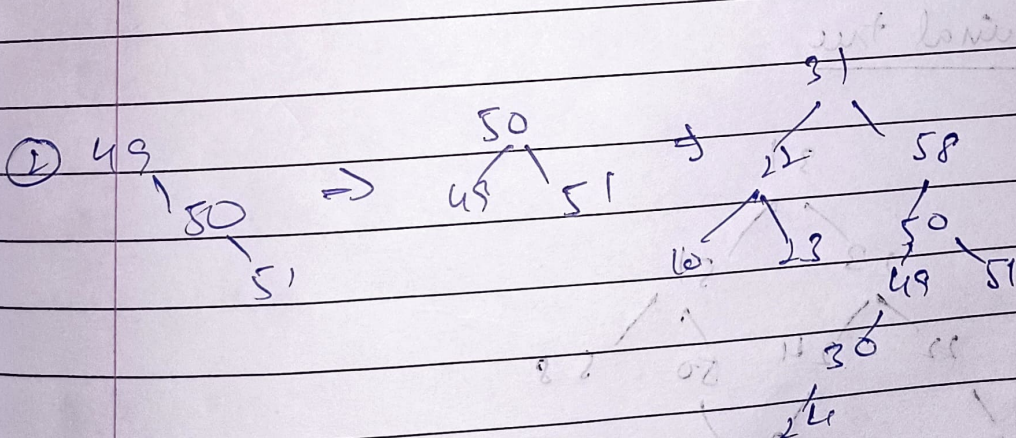
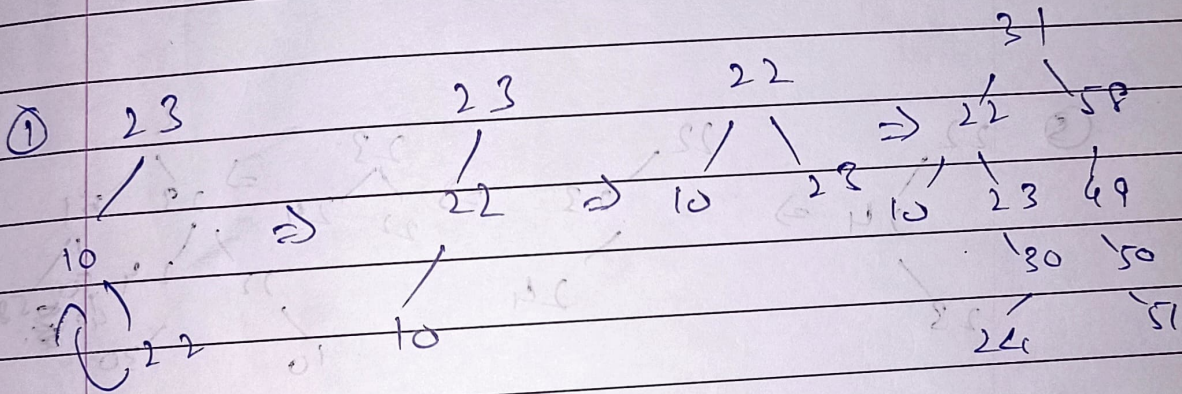
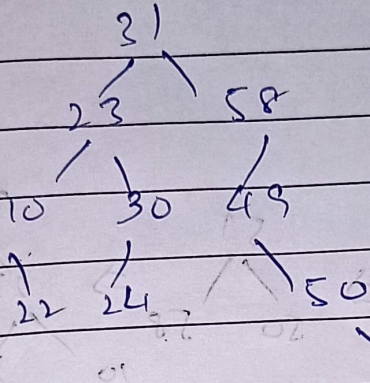
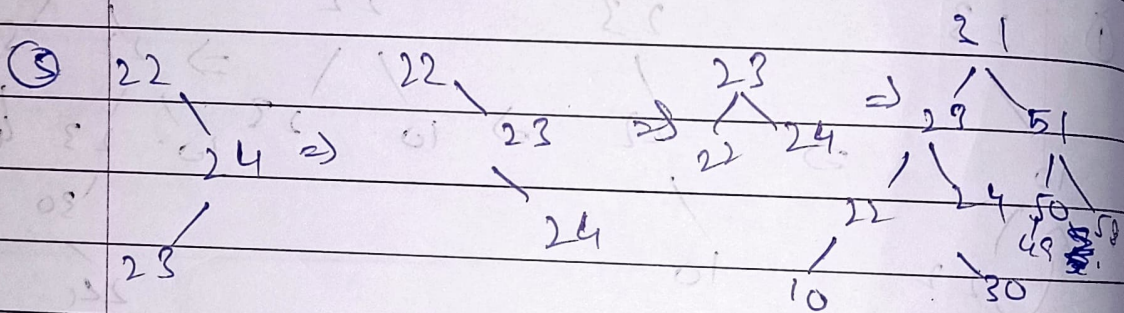
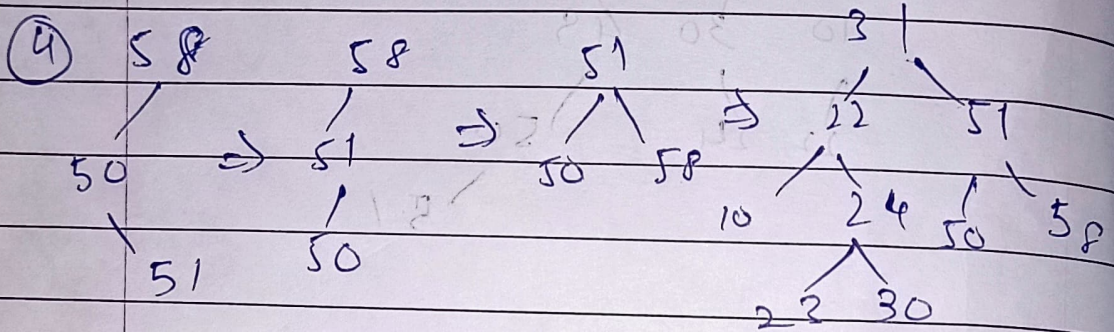
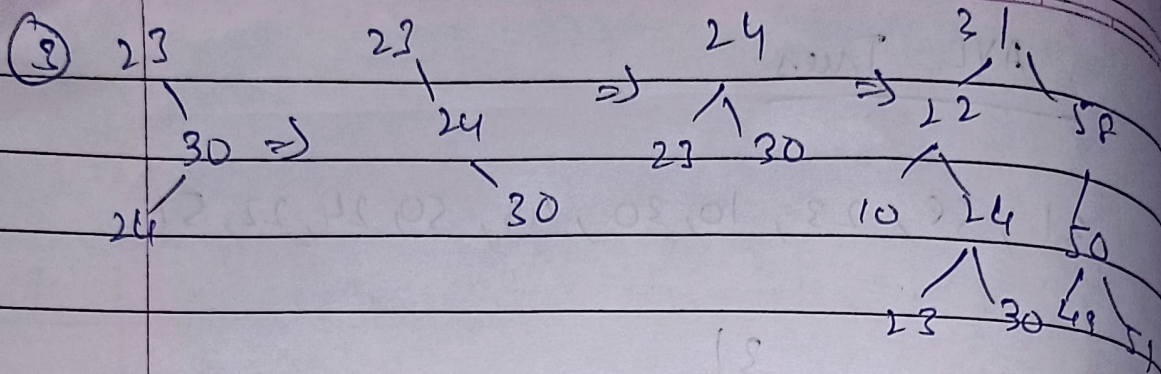


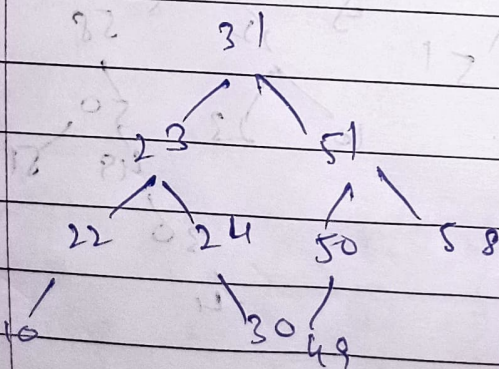
AVL Trees

31, 58, 23, 10, 30, 49, 50, 24, 22, 51





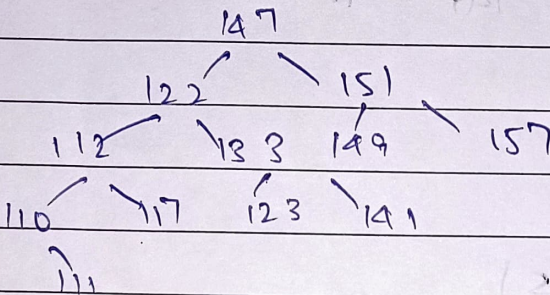
Final tree



Red-Black Tree

Rules

1. The nodes are either Red or Black
2. The root node is always Black
3. All null nodes are Black
4. A red node can't have red child/parent



S1 : 157

S2 : 157

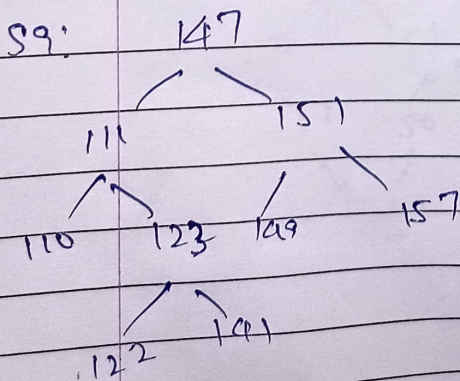
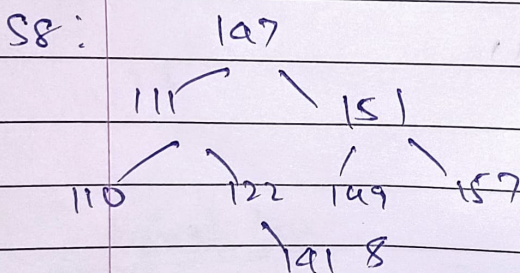
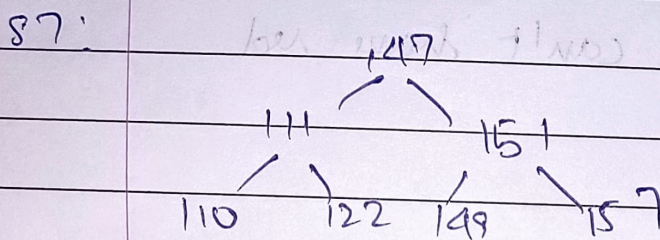
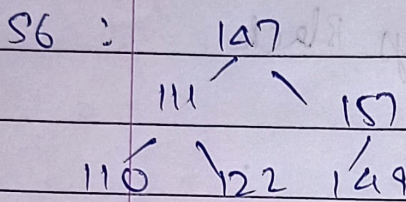
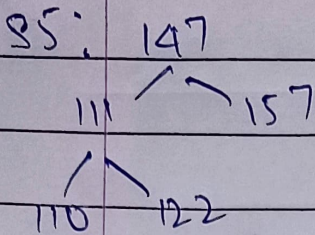
116

S3 : 147

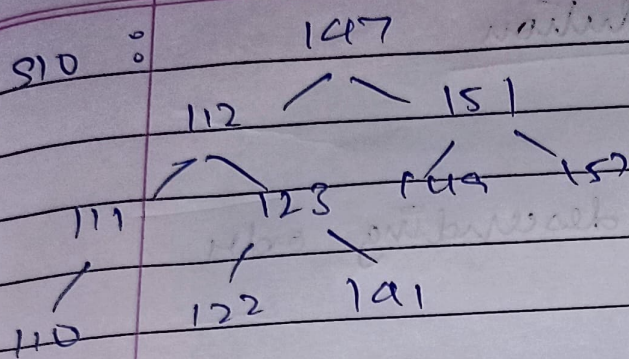
110 157

S4 : 147

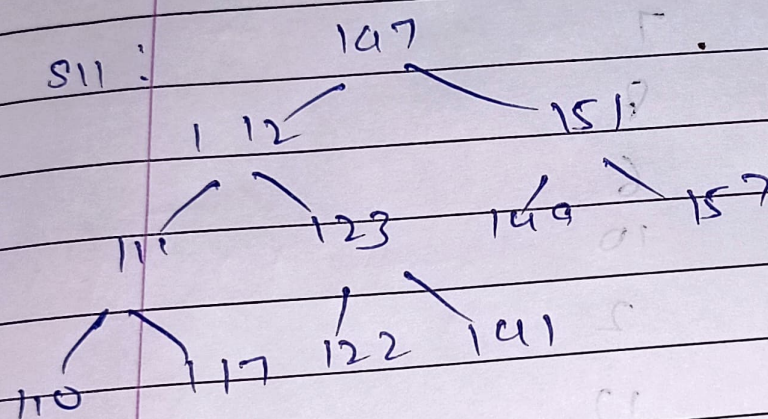
110 157
122



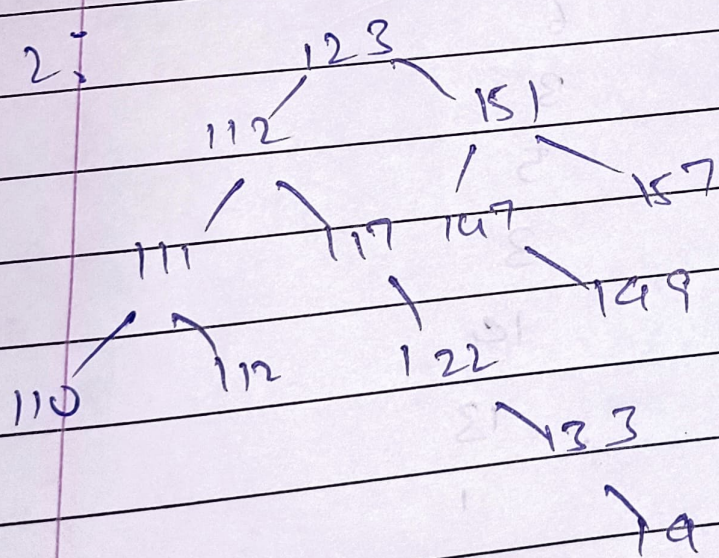
S10 :



S11 :



S12 :



... to ...